

Total No. of Questions : 10]

SEAT No. :

P2136

[Total No. of Pages : 4

[5254]-531

B. E. (Mechanical)

REFRIGERATION AND AIR CONDITIONING

(2012 Pattern) (End Semester)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 4) *Assume suitable data, if necessary.*
- 5) *All questions are compulsory.*

- Q1)** a) In a refrigeration plant working on Bell Coleman cycle, operates between pressure limits of 1.05 bar and 8.5 bar. Air is drawn from cold chamber at 10°C, compressed and then cooled to 30°C before entering the expansion cylinder. The expansion and compression follows the law $p v^{1.3} = C$. Determine the theoretical COP of the system. [6]
- b) What are the advantages and disadvantages of vapor compression cycle over Bell Coleman cycle. [4]

OR

- Q2)** a) A refrigerating system operates on reversed Carnot cycle. The higher temperature of the refrigerant in the system is 35°C and the lower temperature is – 15°C. The capacity of the system is to be 12 tonnes. Determine the following. [6]
- i) COP
 - ii) Heat rejected from the system per hour.
 - iii) Power required
- b) Explain with neat sketch 'Evaporative Coolers'. [4]

P.T.O.

- | Temperature(°C) | Liquid heat
(KJ/kg) | Latent heat
(KJ/kg) | Liquid entropy
(KJ/kg) |
|-----------------|------------------------|------------------------|---------------------------|
| 25 | 298.90 | 1166.94 | 1.1242 |
| − 10°C | 135.37 | 1297.68 | 0.5443 |

- OR

- Q5) a)** 10 cmm air at 37°C DBT, 24 % RH, flow through a desert cooler having an adiabatic efficiency of 75 %. What is the final dry bulb temperature and RH (Relative Humidity), and how much water is required in kg/hr? (USE PSYCHROMETRIC CHART-See page-4, Fig. Q. 5 (a)) [6]

- OR

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- b) Discuss the factors affecting Human Comfort. [4]
- c) Derive an expression of Bypass Factor of coil [6]

- Q7)**
- a) Explain with neat sketch '**Summer Air Conditioning System**'. [6]
 - b) Write note on '**Variable Refrigerant Flow System**'. [6]
 - c) Explain working of Capillary tube and list its advantages and disadvantages. [6]

OR

- Q8)**
- a) Explain with neat sketch '**All Year Air Conditioning System**'. [6]
 - b) Write note on '**Variable Air Volume System**'. [6]
 - c) Explain with neat sketch '**Evaporative Condensers**'. [6]

- Q9)**
- a) Explain Equal Friction Method of Duct Design. List its advantages and disadvantages. [5]
 - b) A rectangular duct of $0.15 \text{ m} \times 0.12 \text{ m}$ is 20 m long and carries standard air at the rate of $0.3 \text{ m}^3/\text{s}$. Calculate the total pressure required at the inlet of the duct in order to maintain this flow and the air power required. Take friction factor, $f = 0.005$. [6]
 - c) Write note on **Fan Laws**. [5]

OR

- Q10)**
- a) Write a note on Classification of ducts. [5]
 - b) Explain with neat sketch '**Fan Coil Unit**'. [5]
 - c) A circular duct of 40 cm diameter is used to carry air in an air conditioning system at a velocity of 440 m/min. If this duct is to be replaced by a rectangular duct of aspect ratio of 1.5, find out the size of rectangular duct for equal friction method. [6]

When

- i) Velocity of air in two ducts is same.
- ii) The discharge rate of air in two ducts is same.

If $f = 0.015$, find out the pressure loss per 100m length of the duct. Take the density of air = 1.15 Kg/m^3 .

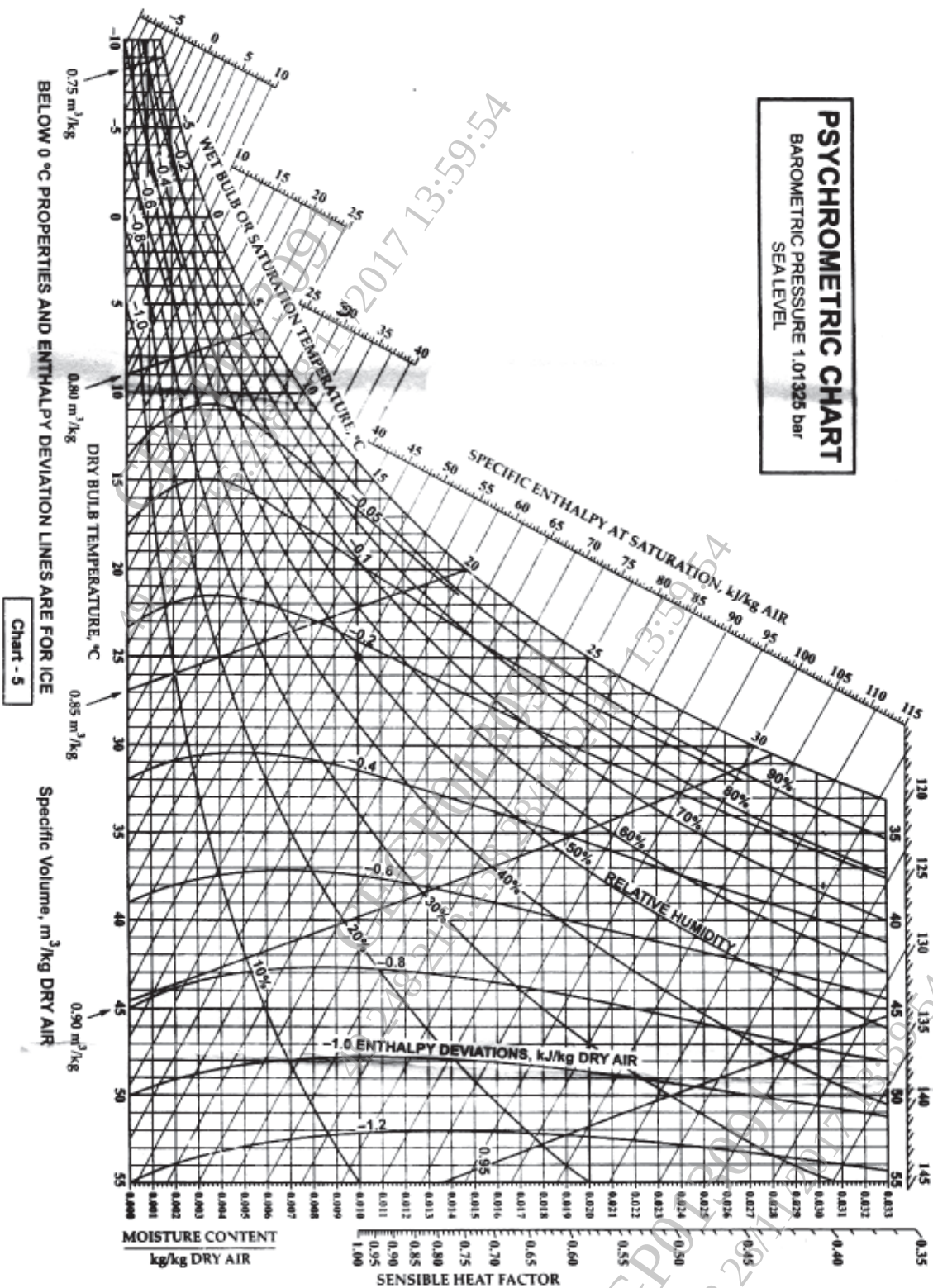


Fig. Q.5 (a) - Psychrometric chart

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